Prepared for referee review

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

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Revision R79 responds to the 11 October 2026 advisory report on R67. The original Neural Bellman Operators paper retains attained continuation fits and their original-law cost comparisons, and adds sharp deployment allocation with installation charges and finite-cohort implementation. All inherited scientific observations and unfavorable results are preserved.

1. RESPONSE TO THE REFEREE

1.1. *Neural Bellman Operators — Revision R79*

We thank the referee for distinguishing the validity of a numerical certificate from the contribution of a fitted continuation and the economic performance of its implemented controller. We revise the original Neural Bellman Operators paper. Its title, author, investment primitives, original Bellman optimum, constructive continuation and historical applications remain unchanged.

The controlling report is the owner-commissioned advisory report of 11 October 2026 on R67, at commit 4b27b56f4514f6c1d77b69fe707f7202f8b06b19, reviewing manuscript eacd3e217ae1e2e9ea6e154b48da7d103159d051. It is not an Econometric Society editorial decision. R79 continues the latest ordinary R78 manuscript at 6b88c6b3a61193c98c822bd62af1203df09c992d, including the executed R77 evidence at 21a99a92c9a8aed96b36931467a40cd7b82aeeb6. The unfinished R78 publication is not described as a completed verification, and its missing publication manifest is not manufactured.

The intervening revisions now connect an attained own-future continuation fit to the costs of the same complete feasible controller. The retained original-law experiment identifies lower ReLU-controller costs than matched Bernstein-controller costs at both acquisition resolutions. R78 makes that finding explicit and adds a sharp simultaneous choice theorem: all marginal and paired intervals imply attainable

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catalogue-regret bounds and exact price regions. The complete graph of jointly qualified comparisons identifies the cost ordering, not only separate pairwise signs. This is an extension of the original construction-to-economic-decision problem, not a retitled prediction-guided search paper.

The retained pure-choice theorem appears in **Simultaneous choice among constructed controllers**, next to its full proof and exact fee calculations in the Numerical and Development Appendix. The new **Deployment choice under qualified cost uncertainty** theorem is in the main article, with its full allocation, installation-cost and finite-cohort proofs in that appendix. All inherited active section files remain byte-identical and are compiled once. Original R77 and R78 wrappers and responses are preserved. Twenty inherited sharp-choice tests and twenty-two new allocation tests accompany the full inherited mathematical and independent numerical-replay chain. R79 adds no simulation rows and does not retime original scientific services.

1.2. *R79 addition: attained deployment allocation and explicit setup costs*

R79 continues the original R78 manuscript at 6b88c6b3a61193c98c822bd62af1203df09c992d and the same controlling R67 advisory report. The earlier publication attempt did not complete, so its inherited R77 logs are not used as R79 verification. The present build must execute the entire retained verification chain and compile the actual new ordinary sources.

The new sharp deployment-allocation theorem extends the original controller-choice problem. Nonnegative controller shares and a common attaining cost potential reduce worst-case expected catalogue regret to a finite matrix game. Exact primal and dual feasible probabilities certify the solution and its optimization gap. The new proof distinguishes a pre-observation mixture of entire policies from averaging actions, fitted coefficients or information sets. Installation costs are not presumed linear: an installed-subset formulation charges every supported controller and retains outside controllers as oracle comparators. A finite-cohort construction realizes the allocation in expectation without pretending that the statistical sample size is the deployment horizon.

For the two attained ReLU precision levels, the complete cost intervals imply an exact fee-dependent allocation and the additional cost that would erase its advantage. At the ambiguity midpoint, the worst-case expected-regret certificate is half the best pure certificate before extra setup charges. This is a sharp economic decision calculation from existing qualified costs, not a new simulation, neural speed claim, fee calibration or unconditional statistical-risk result. The corresponding theorem, full proof, rational primal-dual witnesses and finite-cohort exhaustive checks are deposited. The new suite has twenty-two unskipped exact tests; the retained cost catalogue supplies seven full-catalogue linear-program checks, thirty-five finite-cohort checks and four explicit setup-charge regimes. These are decision calculations, not new economic observations. Manski (2004) is cited for the established minimax-regret decision perspective.

All R78 active section files remain byte-identical and are compiled once. The pure-choice exposition is placed next to its full proof in the Numerical and Development Appendix; no theorem, proof or unfavorable experiment is deleted. The introduction and conclusion have been rewritten to remove accumulated repetition and distinguish attained continuations from optional query proposals. Their original wrappers and response are preserved. The subsequent ten point-by-point replies remain applicable; the new result further addresses M4/M5/M8/M9 and B3/B4/B7/B9 through explicitly costed deployment rather than unsupported neural timing dominance. The high-dimensional, unknown-law and population-ranking distinctions in those replies remain unchanged.

1.3. *M1 / B1. Identify what the neural continuation contributes*

We retain the referee's distinction between an optional action-query predictor and an actual fitted continuation. Correctness of the safeguarded query procedure is method-neutral. It does not imply a neural computational advantage, and the adverse query-selector results remain in the appendix.

The original NBO continuation now has a separate, executed chain. Qualification-optimal fixed-feature fitting minimizes a verified residual-oscillation account conditional on its actually fitted future. An exact primal–dual calculation certifies the

achieved optimization gap. Backward construction returns feasible acquired actors. The original-law experiment evaluates those same actors, with their identities frozen before the evaluation rows. Matched ReLU-versus-Bernstein cost differences are strictly negative at both acquisition resolutions. The manuscript reports this finite, construction-specific result without inferring dominance over all conventional methods or confusing an exact ReLU encoding with a separate algorithm.

R78 adds the next decision in that chain. Its qualified cost set retains all jointly covered observations. Sharp worst-case catalogue regret is computed for each controller and every declared resource-charge vector. An explicitly feasible cost potential attains each upper difference, so the resulting choice guarantee is not merely a loose union of marginal intervals. The classical difference-constraint algorithm is attributed to existing graph methods; our contribution is the executable, price-uniform decision account for the actual constructed NBO controllers.

1.4. *M2 / B2. Match information, representation budgets and complete costs*

The retained R69/R70 study executes certificate-aware hat-ReLU and equally sized Bernstein action readouts, label-loss ReLU and quadratic proposals, alongside prediction-free controls. The learned arms purchase the same declared training information. Exact objective and optimization-gap records accompany the results. Complete cold-start timings include training and retain the unfavorable neural outcomes.

The attained continuation experiment is a distinct matched comparison: ReLU and Bernstein each have nine coefficients, the same coefficient limits, primitive information and acquisition contract at a fixed cover. The actual fitted futures are propagated backwards. R78 compares the resulting original-law controller costs, not action-label fitting objectives or a fixed-label ablation that was never propagated into a separate controller. The two acquisition resolutions remain distinct treatments. Feature matching is not represented as equality of function classes.

The new arbitrary-charge theorem permits construction, acquisition, qualification, loading and installation charges to remain distinct. Its zero-charge ordering is a

statement about expected policy costs only. No missing or shared expense is silently assigned zero to claim a complete-work victory.

1.5. *M3 / B5. Prospective validation and attained training*

The retained readout validation uses predictor-independent contexts disjoint from fitting and executes the prescribed recovery after a failed screen. Its independent-context work bound is not assigned to recursively visited policy states. Predictable qualification and fixed-controller evaluation retain their separate targets.

The later continuation theorem and exact primal–dual calculations address attained fitting rather than the existence of an optimizer. Its qualification objective enters the uniform original-optimum and cost-observation accounts. The current cost experiment evaluates the same frozen complete controllers under independent original-law rows. It does not refit them after observing costs.

R78 is deterministic postprocessing of the existing thirty-estimand simultaneous event. Choosing a cost-minimizing controller, intersecting the two estimators' implications, or examining resource prices after viewing the intervals requires no new independence claim: once every interval covers, the sharp-choice inequality holds for all prices. This does not authorize adding new controllers, new cost estimands or a changed deployment law without qualification.

1.6. *M4 / B3. Prospective reuse and amortization*

The sixty-four-block R69 reuse catalogue retains its predetermined invalidation and stable-crossing rule. The subsequent fresh, exact-key and transported-witness comparisons preserve first fills, failed lookups and actual descendant costs. The executed bounded cache is not said to satisfy stronger retention or fill-precision hypotheses that it did not enforce.

The new result prices actual deployment rather than treating sample size as a reuse horizon. If a per-deployment cost gain is in [l,u], then m deployments and monetary scale v give a gain in [mvl,mvu] before incremental charges. R78 computes exact controller-choice and acquisition-fee regions from all qualified costs. Actual

repeated-use timing evidence remains separate; the theorem does not manufacture an unobserved stable amortized neural speed advantage.

1.7. *M5 / B4. Resolve actual policy costs at the declared precision*

The earlier wide R67 intervals are preserved. The intervening centered evaluation meets its equivalence margin for the callable recursive policies but does not identify strict ordering. Those policies are not the attained continuation actors evaluated in the later original-law experiment.

For the latter, the frozen design uses 524,288 rows, forty-bit continuous bins, five controllers, and two estimators: complete cash cost and exact last-innovation conditional integration. All five marginal costs and ten paired differences for each estimator enter one thirty-estimand family. Every paired interval meets the declared full-width target of 1/256. Only four of twenty satisfy the separate equivalence criterion, because other narrow intervals identify material differences. The width account retains numerical enclosure, endpoint dispersion and finite-support terms.

R78 uses the already jointly covered intervals to identify the full cost order: fine ReLU, coarse ReLU, fine Bernstein, coarse Bernstein, then the installed zero controller. At the two matched resolutions, the Bernstein-minus-ReLU gains are enclosed by [0.0758058,0.0775680] and [0.0753723,0.0771589], respectively, with outward printed endpoints. The exact record is authoritative. These are actual complete-policy comparisons, not differences between qualification certificates.

Both estimators refer to the same costs and share the same coverage event. Intersecting their linear implications adds no sample size. Transitive closure does not further tighten the paired endpoints in this particular dataset; the revision does not claim a new precision improvement where none occurs. Its new contribution is an attainable decision bound and exact selection regions using the complete paid information.

1.8. *M6 / B6. Vector comparison and horizon costs*

The retained vector theorem, trained two-output proposals and conventional-comparator catalogue remain intact. The four-date readout is still labeled as trans-

fer, not four-date supervised fitting. The large simplicial descendant tree and all witnessed-reuse misses retain their original work records.

The actual attained-controller cost experiment has two states, two dates and scalar control. R78 does not relabel it as a high-dimensional or long-horizon comparison. The sharp-choice theorem applies to any finite set of qualified complete controllers, including vector controllers when their cost intervals have actually been obtained. That mathematical applicability does not create a new vector execution. A broader trained long-horizon complete-work advantage is not established by the present cost ordering.

1.9. *M7 / B7. Independently reimplement the critical numerical calculations*

The original-law cost experiment has two separate full-bin implementations. One uses outward binary64 arithmetic on centered primitives. The other uses directed integer endpoints on a dyadic grid, expanded polynomials and independently enumerated acquisition rectangles, importing none of the production economic or actor routines. Both evaluate every new bin for every controller. The published endpoints use their hull, intersected with the proved support, rather than an optimistic intersection. Additional exact-Fraction trajectories and boundary tests complement the full-bin replay.

The R79 release procedure repeats that full dual-arithmetic calculation and recomputes all statistical intervals. The new decision calculation is independently implemented using all-pairs Floyd closure and repeated single-source Bellman–Ford relaxation. Exact cost potentials are checked against every original constraint and attain every claimed upper difference. Negative cycles cause failure; they are not repaired by widening selected intervals. Exact fee-boundary and minimax-tie tests exercise the economic conclusions.

This fully covers the critical new constructed-controller cost and choice calculations. It does not independently regenerate every descendant of the historical eight-million-query vector computation. That distinction and the earlier aliasing counterexample remain explicit.

1.10. *M8 / B8. Distinguish return, fitting and timing reliability*

Safe return remains a property of the verifier, not evidence of reliable hidden-layer fitting. The attained fixed-feature optimization certificates are checked independently of solver success messages. The original fitting warnings, finite-seed catalogue and adverse timings are preserved.

R79 measures its current verification, graph-processing and allocation work separately. These clocks do not replace the original qualification service or become a new timing-distribution experiment. The graph bound counts rational additions and comparisons; it does not omit rational bit complexity or the work required to obtain the input intervals. Price-uniform choice concerns specified charges, not uncertain population runtime estimates.

1.11. *M9 / B9. Make the economic decision explicit*

The installed zero-investment reference supplies a replacement decision within the original economy. It is not substituted for stronger conventional query baselines. Matched attained ReLU-versus-Bernstein comparisons and gains over the installed reference are separate results.

The new theorem gives the exact maximum catalogue regret implied by the retained cost intervals. Minimizing this quantity uses precise paired information instead of only marginal upper costs. It yields a controller choice for every resource-charge vector on one coverage event. The attaining-potential proof states exactly the sense in which the bound is sharp: relative to the retained interval inequalities, not relative to all information in the raw observations or all possible statistical procedures.

For acquisition refinement, the identified gain from coarse to fine ReLU is approximately [0.0002277949,0.0011900893]. When only the finer controller bears an incremental per-deployment charge f, its worst-case regret is (f-l)+ and that of the coarse controller is (u-f)+. The fine controller is universally optimal below l, the coarse one above u, and the regret-minimizing choice changes at (l+u)/2, approximately 0.0007089421. All thresholds and boundary choices are calculated with exact

fractions. The other three constructed controllers remain dominated in this stated charge experiment.

Normalized loss units do not identify a monetary scale, an installation fee or a deployment horizon. The manuscript makes these inputs explicit rather than selecting favorable prices to infer a universal purchase recommendation. It now supplies an identified complete-controller comparison and a full conditional choice region within the original economic model.

1.12. *M10 / B10. Keep the paper's subject and preserve substantive content*

We do not retitle the paper or replace its original continuation-construction problem with a generic query-search framework. The active article follows the same NBO chain from construction and qualification through feasible implementation, original-law evaluation and economic choice. The new theorem has its full proof and exact decision records in the Numerical and Development Appendix.

All R77 active section files are compiled byte-identically once across the four-document package. Their earlier R73/R72/R70 preservation checks remain in the build. Original R77 wrappers and response are retained separately. Controlled-diffusion, recursive-preference, temporal-self and game applications retain their own historical hypotheses; the new expected-cost choice theorem is not silently extended to different nonlinear targets. The Econometrica class, author–year references and explicit mathematical premises remain unchanged.

1.13. *Verification and renewed review*

The release distinguishes inherited scientific execution, full independent cost replay, twenty inherited exact choice tests, twenty-two new allocation tests, ordinary-source compilation, clean archive reproduction and actual PDF inspection. The final release record, rather than this response, determines which checks have passed. No new simulated rows, model fits or scientific timing repetitions are claimed in R79.

The positive result submitted for renewed review is the original NBO paper with an attained continuation-to-controller comparison, strictly identified matched costs, sharp simultaneous priced-choice guarantees, and attained deployment allo-

cation with explicit installation costs and finite-cohort implementation. The adverse recursive-query work, unresolved long-horizon scaling and absence of monetary calibration remain separate evidentiary distinctions, not grounds for discarding the original mathematical and economic development.